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Solving Portfolio Optimization Problems using Adiabatic Quantum Computing

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ABSTRACT

This study explores the application of adiabatic quantum computing to portfolio optimization, a critical problem in finance. By formulating the problem as a Quadratic Unconstrained Binary Optimization (QUBO) model, the study integrates constraints such as expected return, transaction cost, and Environmental, Social, and Governance (ESG) scores. Simulations using D-Wave's quantum annealer demonstrate the feasibility of optimizing portfolios comprising hundreds of assets with reduced computational time compared to classical methods. While current quantum devices face limitations in precision, this work lays the foundation for leveraging quantum technologies in financial optimization.

KEYWORDS

Adiabatic quantum computing, Portfolio optimization, Quantum computing, Quadratic Unconstrained Binary Optimization.

INTRODUCTION

Portfolio optimization is a cornerstone of financial decision-making, enabling investors to allocate assets effectively while managing risk and maximizing returns. Traditional approaches, such as the Modern Portfolio Theory (MPT) introduced by Markowitz [1], have relied on classical computational methods to solve these problems. However, as the complexity and size of investment datasets have grown, these methods face significant limitations, particularly in their ability to efficiently solve multi-constrained problems or incorporate emerging considerations like Environmental, Social, and Governance (ESG) factors [2].

Recent advancements in quantum computing offer a paradigm shift in tackling such challenges. By leveraging the principles of superposition, entanglement, and tunneling, quantum computing has demonstrated potential in solving combinatorial optimization problems more efficiently than classical approaches [3], [4]. Quantum annealing, a specialized quantum computing approach, is particularly well-suited for optimization tasks. It enables the exploration of large solution spaces through energy minimization techniques, making it an attractive option for portfolio optimization [5], [6].

This work builds on prior research demonstrating quantum annealing's application in financial contexts. For example, studies using D-Wave quantum annealers have successfully optimized portfolios with up to 60 assets, showcasing the technology's ability to manage high-dimensional datasets [7]. However, the incorporation of ESG constraints—a critical factor in today's investment strategies—remains underexplored. ESG considerations add a layer of complexity to the optimization process, requiring algorithms that balance financial performance with sustainability metrics [2].

The current study aims to address these gaps by developing and testing a Quadratic Unconstrained Binary Optimization (QUBO) model tailored for portfolio optimization. The model incorporates standard constraints such as budget limitations, transaction costs, and market impact, alongside ESG performance metrics. Using synthetic stock data, the research evaluates the efficacy of quantum annealing in identifying optimal asset allocations under these multi-faceted constraints. Additionally, the study benchmarks the performance of quantum approaches against classical and hybrid solvers to assess their relative advantages and limitations.

This investigation not only advances the application of quantum computing in financial optimization but also contributes to the broader discourse on sustainable investing. By integrating ESG factors into a quantum-enabled optimization framework, the study highlights the potential of quantum technologies to address evolving investment priorities in a computationally efficient manner.

METHODS

CONSTRAINT FORMULATION

The proposed QUBO formulation, which addresses the portfolio optimization is divided into a series of constraints, where each constraint addresses a different optimization regime.

To evaluate the correlation between two stock assets, the following risk aversion formula is employed. In this formulation, n_i and p_i represent the quantity and price of asset i, respectively, while σ_{ij} denotes the expected correlation between assets i and j.

The total number of assets is represented by N_a, and γ is a parameter that controls risk aversion. The resulting value, H_0, represents the initial Hamiltonian. This formulation, which aims to penalize stocks that are highly correlated, aligns with the generic QUBO structure:

$$H_0 = \gamma \sum_{i=1}^{N_a} \sum_{j=1}^{N_a} \sigma_{ij}\, p_i\, n_i\, p_j\, n_j \tag{1}$$

The expected investment return constraint aims to reward the model for a high return on investment. The parameter μ_i is the expected return of asset i. Since this constraint aims to reward high-return investments, the formulation will have a negative sign to minimize the overall objective function. This constraint is modeled as:

$$H_1 = \sum_{i=1}^{N_a} \mu_i p_i n_i \tag{2}$$

The concept of budget constraint is to penalize the asset selection for exceeding predefined budget constraint. The expression for this constraint is raised to the power of two, in order to prevent the expression from taking negative values. The constraint is expressed as:

$$H_2 = \rho\left(B - \sum_{i=1}^{N_a} p_i n_i\right)^2 \tag{3}$$

In the above equation, the parameter ρ controls the importance of the budget constraint, while B corresponds to the total budget. The total budget is defined in terms of the number of bits of precision, N_p, used for the integer variables (stock assets). Specifically, N_p defines the maximum value of n_i, which can be expressed as $n_i = 2^{N_p} - 1$. Using this formulation, B is expressed as:

$$B = \frac{2^{N_p} - 1}{2} \sum_{i=1}^{N_a} p_i \tag{4}$$

The following formula describes the cost of the transaction constraint. The parameter $n_{i,0}$ reflects the initial amount of asset i, and δ represents the percentual transaction cost. While the transaction cost depends on the absolute value of the transaction amount, a quadratic dependence is used instead for computational simplicity. Both the absolute volume of the transaction and the quadratic dependence have the same qualitative effect on the formulation. This constraint is included with a positive sign to reflect its punishing role. The constraint is given as:

$$H_3 = \delta \sum_{i=1}^{N_a} (n_i - n_{i,0}) p_i \tag{5}$$

The market impact constraint is designed to reward investments in assets that minimize their impact on the stock market. This constraint is included in the QUBO formulation with a negative sign. The parameter ε quantifies the market impact, and the constraint is expressed as:

$$H_4 = \varepsilon \sum_{i=1}^{N_a} n_i (n_i - n_{i,0}) p_i^2 \tag{6}$$

The purpose of this constraint is to favor stock assets with high environmental, social, and governance (ESG) scores. To achieve this, the ESG score constraint is included in the final Hamiltonian with a negative sign, rewarding assets with strong ESG performance. The parameter ω controls the importance of the ESG score constraint, while e_i represents the ESG score of asset i. This constraint is modeled as:

$$H_5 = \omega \sum_{i=1}^{N_a} e_i n_i \tag{7}$$

COMPLETE HAMILTONIAN

The final Hamiltonian, solving the portfolio optimization problem is a sum of the above presented Hamiltonians representing each of the constraints with their respective signs. As such the final Hamiltonian H can be presented as:

$$H = H_0 - H_1 + H_2 + H_3 - H_4 - H_5 \tag{8}$$

As such, the complete mathematical formulation of portfolio optimization is given by:

$$H = \gamma \sum_{i,j=1}^{N_a} \sigma_{ij} p_i n_i p_j n_j - \sum_{i=1}^{N_a} \mu_i p_i n_i + \rho \left(B - \sum_{i=1}^{N_a} p_i n_i \right)^2 + \delta \sum_{i=1}^{N_a} (n_i - n_{i,0}) p_i - \varepsilon \sum_{i=1}^{N_a} n_i (n_i - n_{i,0}) p_i^2 - \omega \sum_{i=1}^{N_a} e_i n_i \tag{9}$$

DISCRETIZATION

This work utilizes binary encoding due to its favorable properties and the competitive number of variables required to address portfolio optimization problems. Portfolio optimization seeks to determine the parameters n_i that minimize the objective function H. The discrete version of this portfolio optimization problem is solved by transforming the equation into a QUBO formulation using binary encoding for the integer-valued variables (stock assets). The binary encoding is expressed as:

$$n_i = \sum_{s=0}^{N_p} 2^s x_{i,s} \tag{10}$$

In this formulation, the total number of binary variables, which also corresponds to the number of qubits required to solve the problem using a quantum annealer, is given by $N = N_a N_p$. The formulation is restricted to long positions, as the n_i variable (representing stock assets) is limited to positive integers. However, future extensions of this approach could include negative values, allowing for short positions. The proposed QUBO formulation is then simulated using the D-Wave annealer and D-Wave simulators with the datasets and portfolio optimization parameters described below.

DATA GENERATION

Stock data used in the simulations was synthetically generated to evaluate the performance of the algorithms and demonstrate the capability of quantum annealers to optimize large datasets with multiple constraints. The data generation process was designed to closely emulate real-world values for the parameters used in the simulations. All data was uniformly distributed. For instance, ESG scores ranged from 0 to 1, while stock prices generally fell between 0 and 500, with a bias toward lower values. Expected returns (μ_i) and expected correlations ($\sigma_{i,j}$) were represented as single values, emulating average annual values. Asset prices (p_i) corresponded to either current market values or values recorded on the last day of the simulated period. The budget (B) was modeled according to the formula described in the previous section. The generated dataset was saved in a .csv file for further use in quantum annealing simulations.

For simulation purposes, a predefined number of stock assets was randomly selected from a database of 2,000 synthetically generated assets. This random selection process was employed to create diverse datasets for testing the quantum portfolio optimization model.

INITIATION OF OPTIMIZATION PARAMETERS

The initial amounts of assets, $n_{i,0}$, were chosen by randomly selecting a uniformly distributed integer within the range:

$$n_{i,0} = [0, 2 \cdot (2^{N_p} - 1)] \tag{11}$$

The total number of investments to be allocated at each time step, K, was set to 10,000 for all annealer runs. The number of precision bits used for the integer variables (stock assets) was determined as $N_p = \log_2(10{,}001) \approx 13.288 \approx 14$. This resulted in the total number of qubits for each instance, N, being equal to the product of the total number of assets, N_a, and the number of precision bits, N_p.

The parameter values controlling the constraints were set as follows: $\gamma = 1$, $\rho = 6$, $\delta = 2$, $\varepsilon = 2$, and $\omega = 6$. The total number of assets, N_a, was varied between portfolio optimizations to assess the corresponding optimization times. For each portfolio optimization instance, the pool of assets was randomly selected from a predefined selection of N_a assets.

HYBRID SOLVERS

The performance of hybrid quantum-classical solvers provided by D-Wave Systems, specifically the Binary Quadratic Model (BQM) and the Constrained Quadratic Model (CQM), were tested and evaluated in this portfolio optimization study.

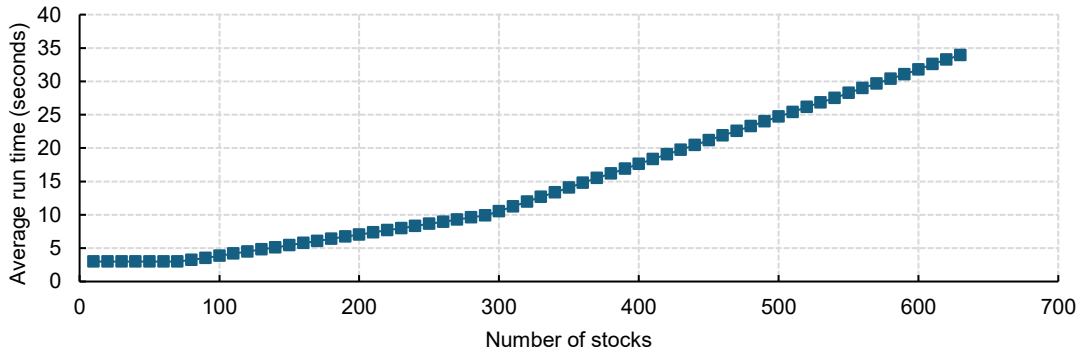
RESULTS

PERFORMANCE OF THE QUBO FORMULATION FOR PORTFOLIO OPTIMIZATION

This study evaluates the performance of D-Wave's hybrid quantum-classical solvers, specifically the BQM solver and CQM solver, using a QUBO formulation for portfolio optimization problems.

The first evaluation involved the performance of the BQM solver available in D-Wave's Leap platform. As hybrid solvers leverage both quantum and classical hardware, they are capable of addressing complex, multivariate problems. Figure 1A illustrates the average run time for the BQM solver across portfolios with 10 to 630 assets. The run time remains relatively consistent for portfolios up to 100 assets, increases linearly up to 300 assets, and then grows more steeply. Each run was repeated three times, and the average values were plotted. Total processing time for generating the results in Figure 1 was 2,763.166 seconds, with the QPU processing time contributing only 10.24 seconds. For the largest portfolio (630 assets), the average optimization time was less than 34 seconds, but total processing—including data preparation, submission to the QPU, and post-processing—required multiple hours. The number of variables in the QUBO model for 630 assets (N_a) and bit dept (N_a) of 14 was 8,820.

A



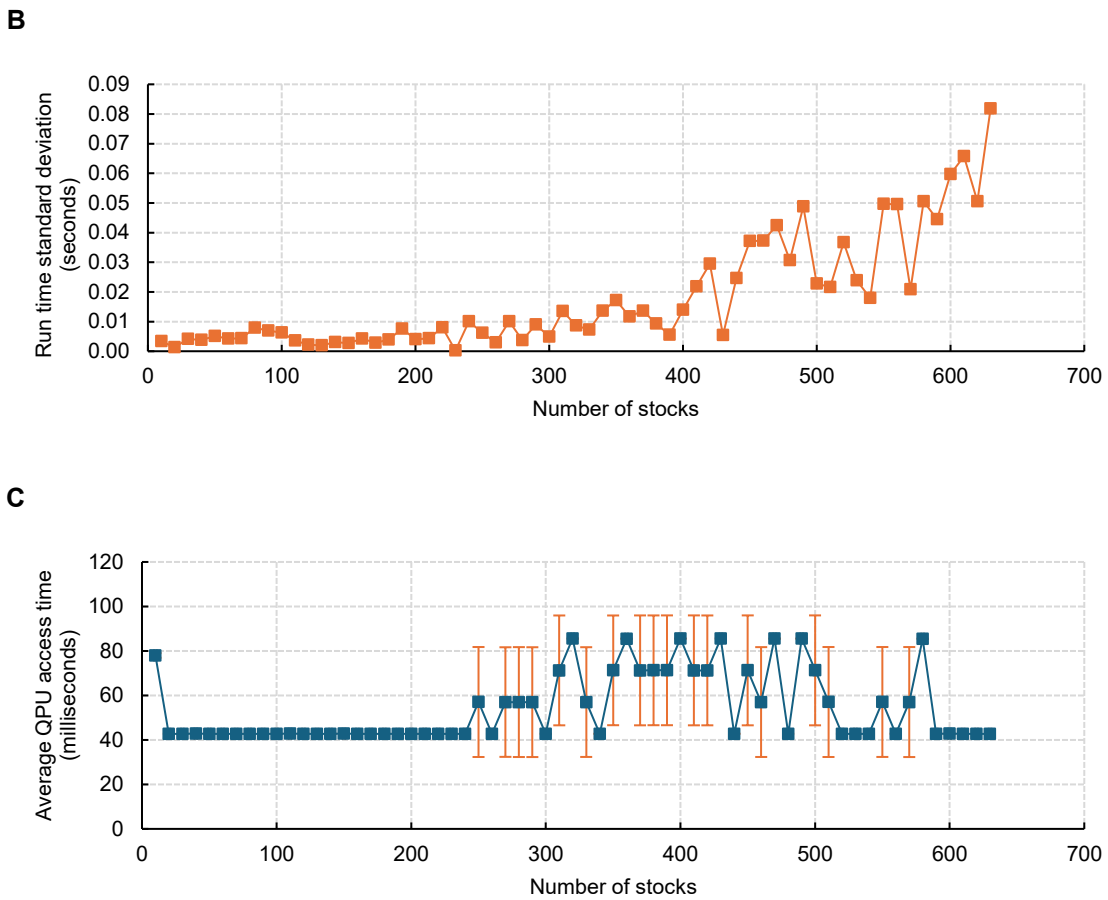


Figure 1. Evaluation of D-Wave Systems binary quadratic model solver. (A) An average run time as a function of the number of portfolio stocks, and (B) corresponding standard deviation. (C) Average QPU access time and corresponding standard deviation. All runs have been repeated three times to calculate average and standard deviation values.

Figure 1B shows the standard deviation of run times across the three repetitions, which is low and consistent for smaller portfolios but increases for larger portfolios, becoming noisier. Similarly, Figure 1C demonstrates the average QPU access time, which remains minimal (measured in milliseconds), even for large portfolios. This highlights that most of the computational workload lies in classical pre- and post-processing, underscoring the efficiency of the QPU for rapid optimization.

For larger portfolios of more than 630 assets, hardware lacked sufficient computational power to compile the QUBO formulations, limiting the analysis. This emphasizes the importance of robust classical infrastructure to support quantum optimization.

MULTI-PERIOD PORTFOLIO OPTIMIZATION

To further evaluate the BQM solver, portfolio optimization was extended to a multi-period scenario involving 10-time intervals (Figure 2). The number of assets was reduced proportionally to maintain similar computational complexity, with portfolios ranging from 10 to 68 assets. Results were consistent with the single-period evaluations, showing linear scaling of run time with two distinct slopes (Figure 2A) and similar trends in standard deviation

(Figure 2B). However, QPU access time increased slightly for multi-period optimization, with greater consistency across repetitions (Figure 2C).

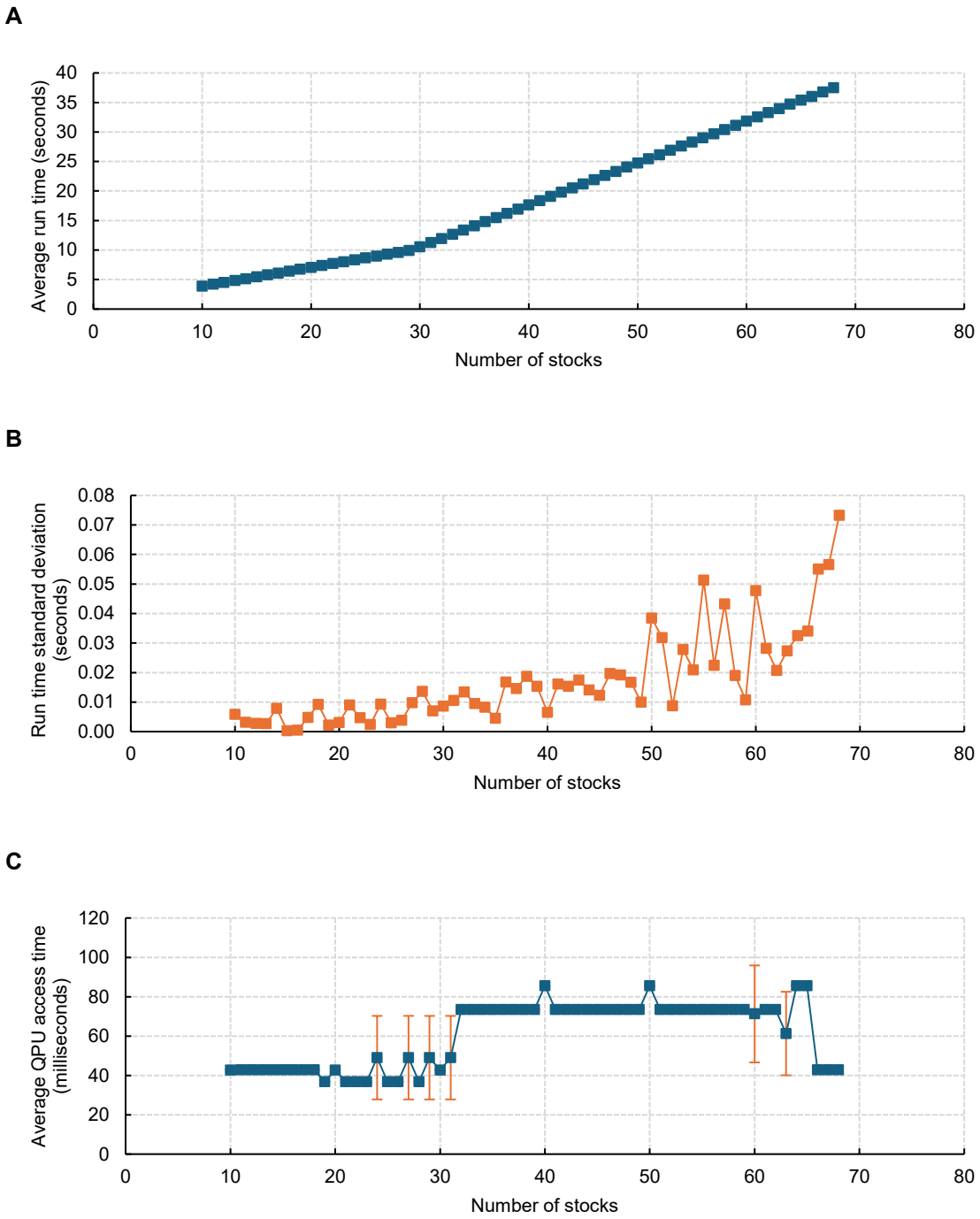
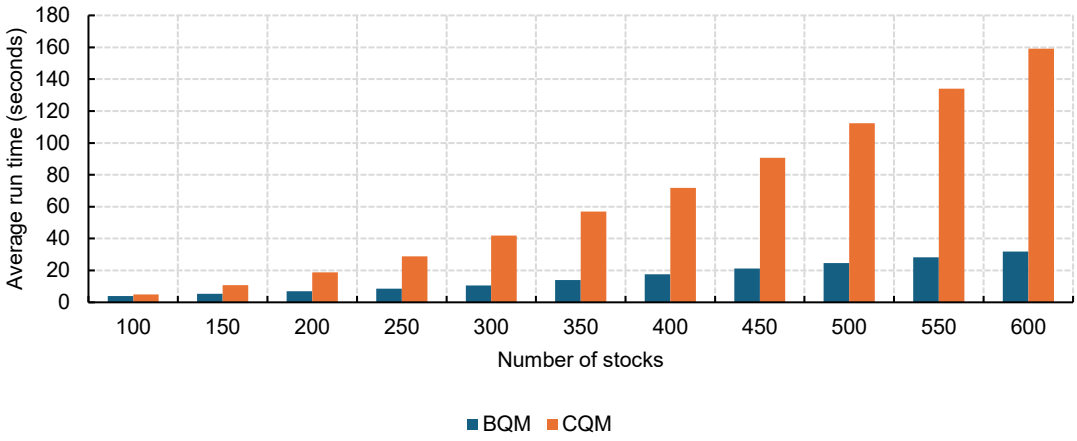


Figure 2. Evaluation of D-Wave Systems binary quadratic model solver for scenarios where portfolio is optimized over multiple periods. (A) An average run time as a function of the number of portfolio stocks, and (B) corresponding standard deviation. (C) Average QPU access time and corresponding standard deviation. All runs have been repeated three times to calculate average and standard deviation values. Portfolio optimization was performed over 10 periods (i.e., days, months).

BENCHMARKING BQM AND CQM SOLVERS

To compare solver performance, the BQM and CQM solvers were benchmarked across portfolios of 100 to 600 assets (Figure 3). The CQM solver required significantly more time to converge, with a fivefold increase in run time for portfolios with 600 assets compared to the BQM solver (Figure 3A). Standard deviation in run time was also higher for the CQM solver (Figure 3B), indicating less consistency across runs. In contrast, the QPU access time was lower and more consistent for the CQM solver than for the BQM solver (Figure 3C).

A



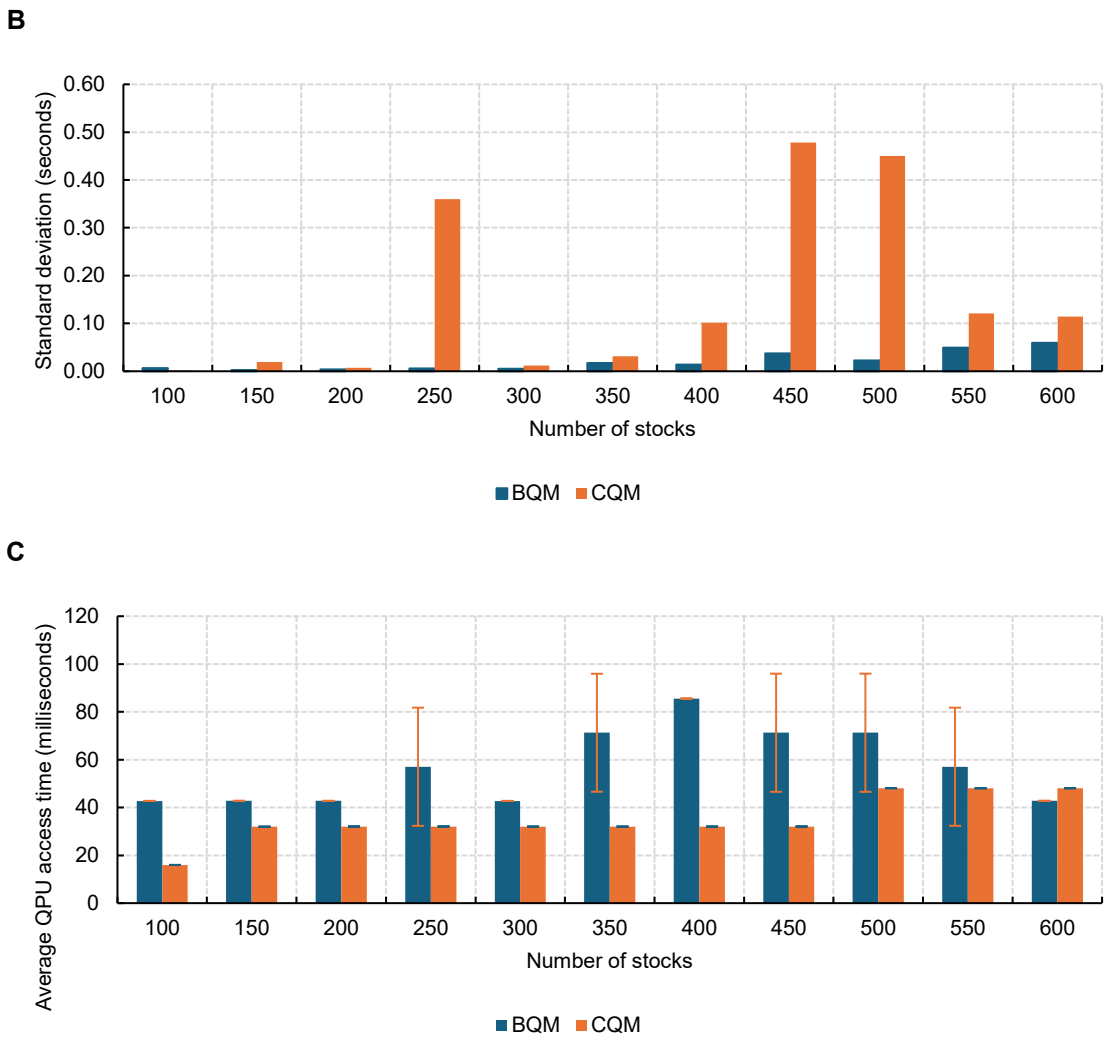


Figure 3. Benchmarking of binary quadratic model solver and constrained quadratic model solver. (A) Average run time as a function of the number of stock assets (variables), and (B) corresponding standard deviation. (C) Average QPU access time as a function of the number of stocks. All runs have been repeated three times to calculate average and standard deviation values.

SOLUTION ACCURACY

To assess solution accuracy, energy values obtained by both solvers were compared for a portfolio of 100 assets (Figure 4). The CQM solver produced solutions with better consistency but lower accuracy, as indicated by a broader range of energy values compared to the BQM solver. The energy range spanned from approximately -420 to 980, suggesting that both solvers face challenges in finding optimal solutions due to the complexity of the QUBO formulation and the large number of constraints.

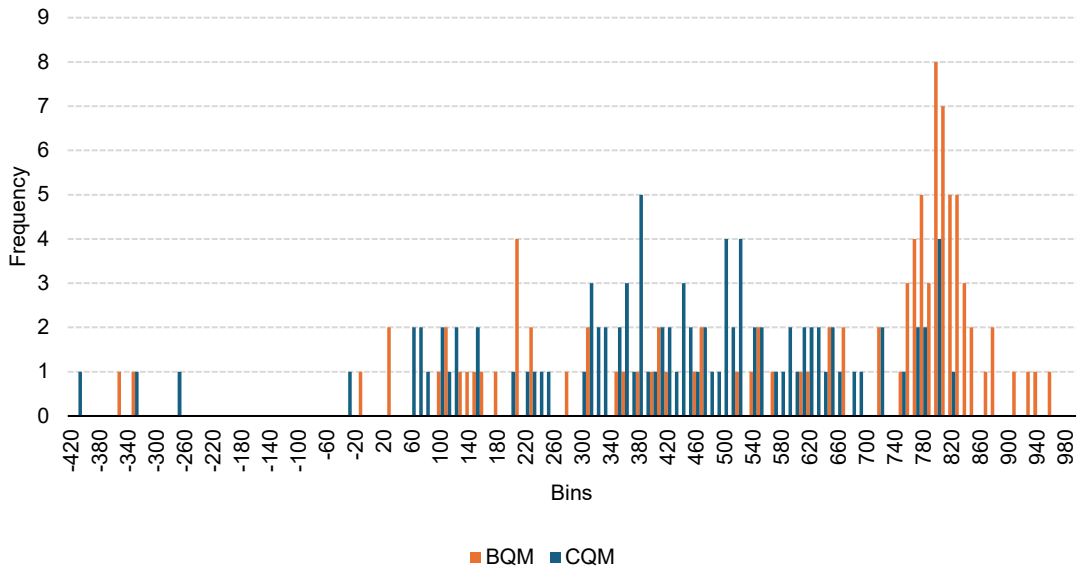


Figure 4. Histogram of solution energy frequency for D-Wave's binary and constrained quadratic model solver. A stock portfolio comprising 100 stock assets has been rerun 100 times for each of the solvers.

The total stock return from the solvers was further compared to results from D-Wave's exact solver for a smaller portfolio of 30 assets (Figure 5). BQM solutions were generally closer to the exact solver's results, with outcomes more symmetrically distributed around the optimal values. In contrast, CQM solutions were more dispersed and often skewed toward lower values.

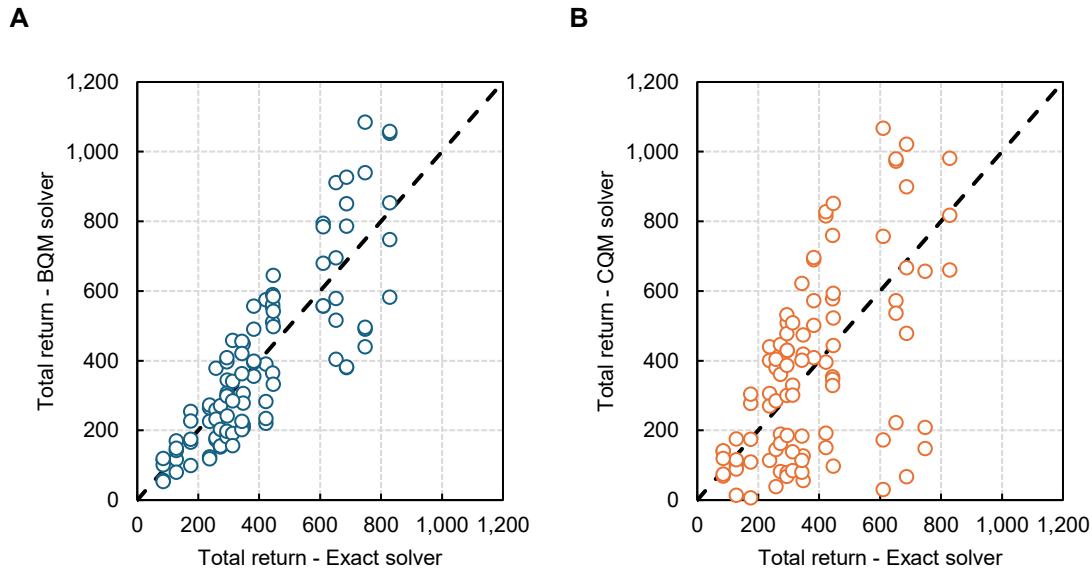


Figure 5. Accuracy comparison. Portfolio returns obtained from Exact solver versus portfolio returns for (A) binary quadratic model solver, and (B) constrained quadratic model solver.

OBSERVATIONS AND IMPLICATIONS

The results demonstrate that hybrid quantum-classical solvers require minimal QPU processing time, with most computational effort expended on classical pre- and post-processing. While BQM solvers are faster and more accurate than CQM solvers in the scenarios evaluated, both exhibit limitations in accuracy due to the complexity of the QUBO formulation and constraints.

The findings highlight the potential of hybrid solvers for portfolio optimization, particularly when classical and quantum approaches are combined to leverage their respective strengths. Further investigations into energy values, ESG scores, and additional constraints are necessary to gain deeper insights into solver performance and applicability for real-world scenarios.

DISCUSSION

This study highlights the potential of quantum annealers in tackling large-scale, multi-constraint portfolio optimization problems. While current devices face hardware limitations, the observed trends suggest that advancements in quantum technology could address these challenges. Incorporating ESG scores into the optimization framework represents a novel contribution, reflecting the growing importance of sustainable investing. Future research could explore real-world datasets, enhanced quantum error correction methods, and expanded constraint sets to improve model applicability and accuracy.

The results of this study underscore the potential of quantum annealing in addressing the increasingly complex demands of financial portfolio optimization. By formulating the problem as a QUBO and integrating constraints such as ESG metrics, budget limitations, transaction costs, and market impact, this work extends the applicability of quantum computing beyond the traditional objectives of risk and return [8].

Previous studies have demonstrated the feasibility of quantum computing in optimization problems, including early experiments with D-Wave systems for portfolio optimization [9], [10], [11]. Cohen et al. showed that portfolios with up to 60 assets could be optimized using quantum annealing, revealing significant reductions in computational time compared to classical methods [7]. Similarly, Venturelli and Kondratyev demonstrated the utility of reverse quantum annealing for multi-constrained optimization problems in finance, further establishing the relevance of this technology [12]. However, most of these studies have focused on relatively simplistic constraints and have not accounted for contemporary investment drivers such as ESG performance metrics.

The incorporation of ESG metrics in this study responds to a critical gap in the literature. ESG considerations are becoming central to investment strategies, driven by regulatory pressures and growing stakeholder demand for sustainable finance [11], [13], [14]. Studies such as those by Chen et al. and Liagkouras et al. highlight the increasing integration of sustainability criteria into financial decision-making, but they largely rely on classical optimization techniques [2], [15]. By demonstrating how quantum annealing can efficiently handle these additional constraints, this work provides a significant advancement in the field.

The findings presented here align with prior evidence suggesting that quantum annealing can outperform classical methods in solving high-dimensional optimization problems [16]. Notably, the scalability of the QUBO formulation to hundreds of assets and multiple constraints is a key outcome, highlighting the practical utility of this approach for real-world financial applications. The results also reaffirm that hybrid quantum-classical solvers, such as those provided by D-Wave, can effectively address the current limitations of quantum hardware by distributing computational tasks.

However, the study also revealed challenges, particularly in accuracy and consistency when compared to exact classical solvers. As noted in similar studies by Preskill, the noisy intermediate-scale quantum (NISQ) era imposes constraints on the precision and scalability of quantum computations [17]. This limitation underscores the need for further development of error mitigation techniques and the enhancement of quantum annealing architectures.

The insights gained from this work pave the way for several future research avenues. First, as quantum hardware evolves, revisiting these methodologies with more advanced systems could yield even greater efficiency and accuracy. Additionally, exploring the integration of more sophisticated constraints, such as real-time market dynamics or regulatory considerations, could further enhance the relevance of this approach. Finally, extending the QUBO formulation to accommodate short positions or other advanced financial instruments would align the model more closely with the needs of institutional investors.

This study contributes to the growing body of evidence that quantum computing can offer transformative benefits for financial optimization. By addressing current limitations and leveraging the unique capabilities of quantum annealing, this work advances both the theoretical understanding and practical application of quantum technologies in finance.

CONCLUSION

This work demonstrates the feasibility of using adiabatic quantum computing for portfolio optimization under realistic constraints, including ESG metrics. By leveraging the QUBO formulation and quantum annealing, the study showcases the potential of quantum technologies to address complex, multi-constrained problems in finance efficiently. While current quantum computing systems are limited by hardware maturity and noise, the results highlight the scalability and computational speed advantages of quantum annealers, offering a glimpse into their transformative potential.

The findings align with the growing body of research advocating for quantum computing's application in finance, but they also reveal existing limitations, such as accuracy and hardware constraints, that must be addressed as the field advances. As quantum computing hardware evolves, its integration into financial systems is poised to drive efficiency and innovation, reshaping portfolio optimization and broader financial decision-making. Future research should build on this foundation, exploring larger datasets, more sophisticated constraints, and real-world data to further refine quantum-enabled financial models.

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DECLARATIONS

ETHICAL APPROVAL AND CONSENT TO PARTICIPATE

Not applicable.

CONSENT FOR PUBLICATION

Not applicable.

COMPETING INTERESTS

The author declares no competing interests.

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